

Geometric Diagnostics for the Virtue of Complexity

Preliminary Grassmann-IPCA Backtest Results, 2020–2025

Internal results note — 24 stocks, 33 characteristics, $\gamma = 0.25$

Preliminary scope. These results are based on a small-universe proof-of-concept: 24 US equities, 33 stock characteristics, and a 5-year window (January 2020 – December 2025). The cross-section is well below the thousands of stocks used in published work, and the sample spans an unusually volatile macro regime. Patterns are *directionally* consistent with theory but should not be over-interpreted as definitive empirical findings.

Background and Framework

We implement the Grassmann-IPCA estimator of Lesniewski & Missaoui [?], which optimises the profiled loss

$$\mathcal{L}(W) = \frac{1}{T} \sum_{t=1}^T \min_f \|r_t - (Z_t W) f\|^2, \quad W \in \text{Gr}(m, k),$$

where $Z_t \in \mathbb{R}^{N \times m}$ is a random-feature characteristic matrix ($\gamma = 0.25$, Rahimi & Recht [?]), W has orthonormal columns (a point on the Grassmann manifold $\text{Gr}(m, k)$), and f is the latent factor.

The diagnostic framework follows Deep et al. [?], who prove that *illusory complexity* manifests as three co-occurring symptoms (Theorem 6.1):

- **Flat Hessian directions** on $\text{Gr}(m, k)$ when k exceeds the true signal dimension k_0 , enabling $O(1)$ subspace rotations under sampling noise.
- **Subspace instability**: large Grassmann distance $d_{\text{proj}} = \|P_t - P_{t-1}\|_F$ across rolling windows.
- **Effective-rank collapse**: $\text{erank}(\hat{\Sigma}_f) \ll k$, because the extra $k - k_0$ factors absorb noise eigenvalues σ^2 rather than signal eigenvalues λ_i .

We vary $k \in \{2, 4, 8\}$, window length $\in \{6, 12, 24\}$ months, and feature count $m \in \{32, 512\}$.

Results

Full results table

Geometric interpretation

Sharpe peaks at $k = 4$, then declines. Across all window lengths and feature counts, Sharpe rises from $k = 2$ to $k = 4$ and then falls at $k = 8$. This is the characteristic complexity curve predicted by Kelly, Malamud & Zhou [?]: additional factors are virtuous up to the true signal dimension k_0 , and illusory beyond it.

Subspace instability grows monotonically with k . d_{proj} rises from ≈ 1.85 at $k = 2$ to ≈ 3.73 at $k = 8$ (for $m = 32$), regardless of window length. This is precisely Theorem 6.1(i) of Deep et al. [?]: when $k > k_0$, the flat Hessian directions allow sampling perturbations to rotate the extra subspace components by $O(1)$ amounts, producing large Grassmann distances between consecutive windows. The signal directions remain stable; it is the noise-absorbing extra factors that spin freely.

γ	m	win	k	Sharpe	avg_stability	avg_erank	collapse_frac
0.25	32	6	2	0.433	1.845	1.515	0.000
0.25	32	6	4	0.464	2.656	2.211	0.340
0.25	32	6	8	0.415	3.734	2.683	0.986
0.25	32	12	2	0.119	1.857	1.471	0.000
0.25	32	12	4	0.309	2.642	2.458	0.218
0.25	32	12	8	0.283	3.726	3.415	0.700
0.25	32	24	2	0.185	1.850	1.402	0.000
0.25	32	24	4	0.280	2.641	2.330	0.307
0.25	32	24	8	0.258	3.719	3.826	0.528
0.25	512	6	2	0.417	1.992	1.617	0.000
0.25	512	6	4	0.432	2.822	2.335	0.272
0.25	512	6	8	0.422	3.984	2.792	0.968
0.25	512	12	2	0.136	1.990	1.643	0.000

Table 1: Rolling-window Grassmann-IPCA backtest, 2020–2025. **win** = estimation window (months); **avg_stability** = mean $d_{\text{proj}} = \|P_t - P_{t-1}\|_F$; **avg_erank** = mean $\text{erank}(\hat{\Sigma}_f)$; **collapse_frac** = fraction of windows where $\text{erank} < k/2$. Bold Sharpe marks the best configuration.

Effective rank collapses for $k = 8$. At $(m = 32, \text{win} = 6, k = 8)$, $\text{avg_erank} = 2.68$ against a nominal $k = 8$, and **collapse_frac** = 0.99: effectively every window is in the collapsed regime. This directly reflects the block-spectral structure of Eq. (26)–(27) in [?]: the population fitted factor covariance has spectrum $\{\lambda_1, \dots, \lambda_{k_0}, \sigma^2, \dots, \sigma^2\}$, so erank collapses toward the true signal count $k_0 \approx 2\text{--}3$ irrespective of k .

At $k = 2$, collapse fraction is exactly zero. When k is at or below k_0 , the subspace is fully occupied by signal directions. The Hessian has no flat directions, d_{proj} is small (≈ 1.85), and erank never collapses — all consistent with the virtuous-complexity regime.

Larger feature sets ($m = 512$) provide limited benefit. Richer random-feature maps ($m = 512$ vs. 32) do not improve Sharpe meaningfully at this cross-section size. Identifiability is likely limited by $N = 24$ stocks rather than the feature dimension, so the additional expressive capacity of the richer map cannot be exploited.

Longer estimation windows hurt at $k = 2$. Sharpe at $(m = 32, k = 2)$ drops from 0.43 ($\text{win} = 6$) to 0.12 ($\text{win} = 12$) to 0.19 ($\text{win} = 24$). This is consistent with a non-stationary signal over the 2020–2025 period: the short-window model adapts faster to the volatility-regime shifts of the COVID/post-COVID era, while the long-window model mixes heterogeneous regimes.

Summary of Geometric Findings

The three hallmarks of illusory complexity — rising subspace instability, collapsing effective rank, and degraded OOS performance — appear together at $k = 8$ and are absent at $k = 2$. The data suggest an effective signal dimension of $k_0 \approx 2\text{--}3$ for this universe and period. These patterns are qualitatively consistent with the predictions of Deep et al. [?] and the empirical findings of Kelly et al. [?], though a much larger cross-section and longer history would be required to draw quantitative conclusions.

Important note: regularisation not yet implemented.

The current implementation does *not* include the Grassmann regularisation term proposed in the forthcoming extension of Lesniewski & Missaoui [?]. Kelly & Malamud [?] show theoretically that with appropriate regularisation, the out-of-sample Sharpe curve exhibits *permanent ascent* — monotonically increasing with k as regularisation prevents the optimizer from wandering into the flat noise directions. Without regularisation, we see the classic *inverted-U* profile (Sharpe peaks at $k = 4$, falls at $k = 8$), indicating we are **not yet in the permanent-ascent regime**. Incorporating the regularisation term is the primary next step, and is expected to change the conclusions for $k = 8$ materially.

References

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